GRADUATE COURSE OUTLINE

Research Methodology

NMEC 595 | Winter 2026

Instructor	Suparno Bhattacharyya
Email	suparno@iitism.ac.in
Office	224A, ME
Office hours	By appointment

Credits	9
Meetings	W/TR/F
Time	15:00–15:50 16:00–16:50 17:00–17:50
Location	MECH-G1

Research design, statistical reasoning, and scientific communication for engineering inquiry.

Course Overview

DESCRIPTION

- Introduces foundational principles of research methodology in scientific and engineering contexts.
- Covers research problem formulation, gap identification, hypothesis development, and study design.
- Includes data collection, interpretation, and statistical reasoning for evidence-based conclusions.
- Emphasizes how to select, justify, and apply methods rigorously in practice.

Course Format

- Lectures and group work
- No prerequisites
- R / Python support
- Scientific and engineering focus

Learning Outcomes I

RESEARCH FOUNDATIONS

Research processes

Identify, design, and execute research problems using scientific and statistical tools.

Methodology selection, logical framing, and structured inquiry.

Measurement tools

Study measurement scales, sources of error, and techniques for developing evaluation instruments.

Measurement rigor and data usability.

Sample design

Understand sample design types, classification, and characteristics of good sampling procedures.

Data collection strategy and sampling quality.

Data quality

Learn methods of data collection together with reliability and validity.

Trustworthy empirical evidence and defensible analysis.

Learning Outcomes II

COMMUNICATION AND INFERENCE

Research communication

Prepare and present structured reports for dissemination of research outcomes.

Clear writing, reporting, and presentation practice.

Statistical reasoning

Use statistical tools for sample design, data analysis, and scientific conclusions.

Inference, interpretation, and justification.

Completion Profile

By the end of the course, students should be able to formulate a research problem, design an appropriate study, collect or simulate relevant data, analyze the results rigorously, and communicate findings in a standard scientific format.

Materials and Software

CORE RESOURCES

1. **C. R. Kothari and Gaurav Garg**, *Research Methodology – Methods and Techniques*, 4th ed., New Age International, 2019.
2. **D. C. Montgomery and George C. Runger**, *Applied Statistics and Probability for Engineers*, 6th ed., 2016.
3. **Ranjit Kumar**, *Research Methodology: A Step-by-Step Guide for Beginners*, 5th ed., SAGE Publications, 2018.

Required Software

R / Python

Instructional Mode

Lectures and guided group work

Assessment and Grading

EVALUATION SCHEME

30%	**50%**	**20%**
Midterm Exam	**End Semester Exam**	**Project Presentation**
MCQ format	Long answer type	Group presentation

Project Component

Students must formulate a research problem, collect or simulate data, apply appropriate statistical tools, interpret results correctly, and justify conclusions using sound statistical reasoning. The final submission includes a structured research report and an oral presentation.

Policies

COURSE POLICIES

Academic Integrity

- All submissions must reflect the student's own understanding and effort.
- AI tools are permitted only when explicitly allowed by the instructor.
- Any AI use must be disclosed clearly.
- Unauthorized or undisclosed AI use constitutes academic misconduct.

Accessibility

Students with documented needs should inform the

Medical / Emergency Cases

Medical or emergency situations require prompt communication and official documentation where applicable.

Communication

- WhatsApp may be used for broadcasting only.
- All formal communication must occur via email.

Key Dates

WINTER 2026 TIMELINE

Midterm	Midterm examination window	**27 Feb – 03 Mar 2026**
Project due	Group project presentation deadline (tentative)	**11–12 Apr 2026**
End-sem exam	End semester examination window	**26 Apr – 03 May 2026**
Holiday / no class	Consult the academic calendar for schedule changes	**See calendar**

CONTACT

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